1. Executive Summary

The Coffee Shop POS is a point of sale system platform developed to be used in daily cafe operations. It addresses the order processing for coffees and desserts, billing, stock management and administration processes. The system was built with JAVA.

2. Product Description and Vision

2.1 Problem Domain

Managing daily operations in a coffee shop has some critical challenges, such as slow order taking, inaccurate inventory tracking, and human error in manual billing can lead to revenue loss and decreased customer satisfaction. Traditional methods like legacy cash registers and spreadsheets greatly increase these risks.

2.2 Solution: Coffee Shop POS

This POS system offers these solutions:

- Order Creation and Customization
- Billing and Payment Processing
- Managing Products
- Real-time Inventory Deduction

2.3 Target Audience

- Independent coffee shops and cafes
- Bakery and pastry shops
- Multi-location cafe franchises
- Any food and beverage organization, because the application is flexible to changes.

3. Core Features and Capabilities

- **Role-Based Access Control (RBAC):** Only the administrator can access the management panel to manage the products in the menu.
- **Rapid Order Processing:** Intuitive, touch-friendly interface designed for speed.
- **Integrated Payments and Receipt Generation:** Supports multiple payment types including cash, card, and digital wallets. Includes customizable digital or printed receipt generation.
- **Sales Analytics:** Tracking daily revenue, and highlighting best-selling items to empower data-driven business decisions.

4. Technical Architecture and Infrastructure

4.1 Overall Architecture

Coffee Shop POS is a client side applications.

- Frontend: Java SE 21
- Backend: Java SE 21

Layered Architecture

Presentation Layer (UI)
Java SE 21 | Java Swing | JOptionPane | Miglayout 11.4.2

Application Layer
Java SE 21 | Encryption Module | Authorization Engine

Backend Layer
JDBC | MySQL Connector/J | Prepared Statements

Database Layer
MySQL | Relational Schemas | Transactional Logging

Infrastructure Layer
GitHub Actions (CI/CD) | Local JVM Runtime (Execution)

4.2 Project Structure

The source code follows a modular and industry-standard folder structure:

- **`src/com/coffeeshop/ui/`** – View components, Layouts, and Swing Forms.
- **`src/com/coffeeshop/security/`** – Encryption modules, Auth logic, and BCrypt utilities.
- **`src/com/coffeeshop/data/`** – DAO (Data Access Objects), Database connections, and SQL scripts.

4.3 Data Flow

When a user attempts to access confidential data:

1. User enters a Username and Password.
2. Username is checked for a match in the system.

3. Password is authenticated.
4. A new screen opens with different options according to the user's access level.
5. The access event is recorded in the audit log.

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5. Technology Stack

5.1 Frontend Technologies

- Java SE 21 – Core runtime and object-oriented programming logic
- JavaX Swing – UI components and event handling
- MigLayout 11.4 – layout engine for UI grids
- JOptionPane –system for user alerts and notifications

5.2 Backend and Database

- MySQL - Relational database management system for structured data storage and ACID-compliant transactions.
- JDBC (Java Database Connectivity) – API for communication between the Java application and the MySQL database.
- SQL (Structured Query Language) – Optimized data definition and manipulation via relational schemas and stored procedures.

5.3 Code Distribution

- Java – 97% (core application logic)
- SQL – 3% (stored procedures)

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6. Security Architecture

Coffee Shop POS follows a defense in depth strategy, implementing independent security controls at every layer.

6.1 Authentication

- Secure Credential Validation.
- Session State Management.
- BCrypt Password Hashing.

6.2 Authorization

- RBAC: SuperAdmin, Admin, Staff
- Clearance Levels: L0 (Public) – L4 (Top Secret)
- Functional Module Restriction – Java Swing UI components are dynamically enabled/disabled based on the authenticated user's permissions.
- Principle of Least Privilege

6.3 Data Security

- Application-level encryption module
- Prepared Statements **(**JDBC**)** – Use of parameterized queries to prevent SQL Injection attacks.
- Local Environment Configuration – Secure management of database credentials and API keys via external properties files.
- Sensitive Data Masking – UI-level masking for passwords and sensitive identifiers within the Swing interface.
- Encryption at Rest – MySQL-level transparent data encryption for sensitive table columns.

6.4 Auditing and Traceability

- Transactional Logging – Database-level triggers and Java-side logs to record all data modifications.
- Timestamped Operation History – Every record includes automated created_at and updated_at metadata.
- User Activity Tracking – Centralized audit table capturing user IDs and performed actions.
- Compliance-ready reporting support (e.g., ISO 27001, data protection regulations)

6.5 Backup and Disaster Recovery

- Full system backup and restore
- Version history for data loss prevention
- Point-in-Time Recovery – Binary log configuration to allow restoration to a specific transaction state.
- Manual System Snapshots – Procedure for full database exports before major system updates.

6.6 Secure Development Practices

- Java Static Analysis – Leveraging Java's strong typing and compile-time checks to reduce runtime vulnerabilities.
- Modular Code Architecture – Separation of concerns between UI and Data.
- Dependency Management – Regular auditing of external JAR files (e.g., MySQL Connector/J) for known vulnerabilities.
- CI/CD pipeline via GitHub Actions
- SECURITY.md vulnerability disclosure policy

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7. Role and Authorization Management Model

The Coffee Shop POS implements a robust **Role-Based Access Control (RBAC)** model to ensure that system functions are only accessible to authorized personnel. This model simplifies security management by linking permissions to specific job functions rather than individual users.

- **Role Hierarchy:**
  - **SuperAdmin:** Possesses full system-wide authority, including access to auditing and disaster recovery configurations.
  - **Admin:** Responsible for managing the management panel, product menus (Drinks and Desserts), and reviewing sales analytics.
  - **Staff:** Primarily focuses on the front-end POS interface, handling rapid order processing and receipt generation.
- **Authentication & Validation:**
  - Users must undergo **Secure Credential Validation** by entering a unique Username and Password.
  - The system uses **BCrypt Password Hashing** to ensure that stored credentials remain encrypted and secure from data breaches.
- **Authorization Engine:**
  - Once authenticated, a dedicated **Authorization Engine** within the Application Layer determines the user's permissions.
  - The system then opens a screen with specific options tailored to that user’s access level.

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8. Clearance Level System

To achieve a **Defense-in-Depth** strategy, the system categorizes data and functions into five distinct clearance levels, ranging from L0 to L4.

Clearance Level	Classification	Description
L0	**Public**	Information accessible to all users, such as general product names.
L1 - L2	**Internal**	Standard operational data required for daily coffee shop sales and billing.
L3	**Confidential**	Business-sensitive data including inventory tracking and historical sales analytics.
L4	**Top Secret**	Highest security tier involving **Transactional Logging**, **Audit Tables**, and sensitive system configurations.

- **Functional Module Restriction:** The system dynamically enables or disables **Java Swing UI components** based on these levels, ensuring users only see what they are cleared to access.
- **Principle of Least Privilege:** Access is strictly limited to the minimum information and tools necessary for a user's specific role.

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9. User Experience and Interface (UI/UX)

The Coffee Shop POS is designed with a focus on speed, reliability, and security, utilizing modern Java-based UI technologies.

9.1 Technical UI Architecture

- **Core Framework:** Built using **Java SE 21** and **JavaX Swing**, providing a stable and high-performance desktop environment.
- **Grid Layout Management:** Uses **MigLayout 11.4.2** to create a sophisticated, organized, and scalable interface grid.
- **Interaction Design:** Employs **JOptionPane** for critical user alerts, notifications, and error handling to ensure a responsive feedback loop.

9.2 Key UX Features

- **Rapid Order Processing:** The interface is optimized for speed and is compatible with **touchscreen monitors**, which are essential for high-traffic cafe environments.
- **Sensitive Data Masking:** To maintain privacy in public shop areas, the UI masks passwords and other sensitive identifiers within the Swing interface.
- **Role-Specific Views:** The UI adapts to the user; for example, the management panel is hidden from regular Staff to prevent unauthorized menu changes.
- **Optimized Resolution:** Designed for a minimum display of **1024x768**, ensuring all menu items and administration panels are clearly visible.

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10. Installation and Deployment

The Coffee Point of Sale System is designed to be installed and deployed across a variety of environments, ranging from single-terminal café setups to multi-location enterprise deployments. This section outlines the recommended procedures for installing the system, configuring the necessary environment, and deploying the application in both development and production contexts.

10.1. Prerequisites

Before installation begins, the host environment must satisfy a set of software and hardware prerequisites to ensure the Coffee POS System operates reliably. All server-side and client-side dependencies described in Section 14 must be fulfilled prior to beginning the installation process.

Required prerequisites include:

- Java Development Kit (JDK) 17 or higher installed on the host machine.
- Apache Maven installed and configured for build management.
- MySQL or PostgreSQL database server running and accessible.
- Docker installed if containerized deployment is preferred.
- Minimum 4 GB RAM and 10 GB available disk space on the server.
- Network connectivity between all POS terminals and the database server.

10.2. Installation Steps

The installation process follows a structured sequence of steps to ensure that all system components are correctly configured before the application is launched. Administrators should follow the steps below to complete a standard installation.

Step	Action	Description
1	Clone or Download Repository	Obtain the Coffee POS source code from the official repository or distribution package.
2	Configure Environment Variables	Set required environment variables including DATABASE_URL, ADMIN_DEFAULT_USER, ADMIN_DEFAULT_PASSWORD, APP_NAME, and DEFAULT_CURRENCY.
3	Initialize the Database	Execute the provided SQL schema scripts to create the necessary database tables for menu items, orders, and user credentials.
4	Build the Application	Run the Maven build command (mvn clean install) to compile the source code and resolve all project dependencies.
5	Launch the Application	Execute the compiled JAR file or use Docker Compose to start the application container in the target environment.
6	Verify Installation	Access the Login Frame using the default administrator credentials and confirm that the POS interface loads correctly.

10.3. Docker Deployment

For organizations that prefer containerized deployment, the Coffee POS System supports Docker-based installation. Docker ensures a consistent runtime environment across development, testing, and production systems, reducing compatibility issues during deployment. A Docker Compose configuration is provided to orchestrate the application container alongside the database service.

Docker deployment steps:

- Ensure Docker and Docker Compose are installed on the host machine.
- Configure environment variables in the provided .env file.
- Execute docker-compose up to build and launch all required containers.
- Access the application through the configured host and port settings.
- Use docker-compose down to gracefully stop all running containers.

10.4. Post-Installation Configuration

After the application has been successfully launched, administrators should complete the following post-installation tasks to prepare the system for operational use.

- Log in to the Admin Panel using the default administrator credentials.
- Update the default administrator password to a secure custom credential.
- Configure the café menu by adding Drinks and Desserts through the Menu Management interface.
- Create additional user accounts for cashiers and POS operators as needed.
- Verify that transaction logging and daily sales tracking are functioning correctly.

11. Integration and Scalability

The Coffee Point of Sale System is architected to support integration with external services and to scale alongside the operational growth of a café or coffee shop business. This section describes the integration capabilities available within the system and outlines the scalability strategies that allow the platform to support businesses from a single-terminal café to a large multi-location chain.

11.1. Integration Capabilities

The Coffee POS System is designed with extensibility in mind, allowing organizations to integrate the platform with third-party services and existing business systems. Integration points are available across several functional areas of the application.

Integration Area	Description	Relevance to Coffee POS
Database Integration	The system supports MySQL and PostgreSQL as backend database solutions.	Allows cafés to connect the POS to an existing database infrastructure or provision a new database as part of the deployment process.
Payment Processing	The system can be extended to support external payment gateway APIs for card and digital wallet transactions.	Enables cashiers to process card and contactless payments directly through the POS interface.

Integration Area	Description	Relevance to Coffee POS
Inventory Management	Integration with third-party inventory tracking systems is supported in the Professional and Enterprise packages.	Allows café managers to monitor ingredient stock levels and receive automated reorder notifications.
Reporting and Analytics	Sales data can be exported and integrated with external business intelligence and reporting tools.	Enables management to analyze sales trends, peak hours, and product performance using preferred analytics platforms.
Authentication Services	The system supports integration with enterprise identity providers for centralized user authentication.	Allows large café chains to manage user credentials through a unified identity management solution.

11.2. Scalability Architecture

The Coffee POS System is built using a modular class-based architecture that supports incremental scaling as business needs evolve. The separation of core system components — including the POS interface, Admin Panel, Menu Management, and database layer — ensures that individual components can be upgraded or expanded without requiring a complete system overhaul.

Key scalability features include:

- Multi-location support: The Professional and Enterprise packages support deployment across multiple café branches with centralized menu and administrative configuration.
- Multiple POS terminals: Additional POS terminals can be connected to a shared database server, enabling concurrent order processing across a single café location.
- User account scaling: The system supports the addition of unlimited user accounts in the Enterprise package, accommodating large staff rosters across multiple locations.
- Containerized deployment: Docker support allows the system to be horizontally scaled by deploying additional application containers as demand increases.
- Menu extensibility: The Menu class is designed to accommodate future menu categories beyond Drinks and Desserts without requiring structural changes to the core system.

11.3. Scalability by Package

The scalability capabilities of the Coffee POS System are directly aligned with the commercial package selected by the organization. The table below outlines the scalability provisions available at each tier.

Scalability Feature	Starter	Professional	Enterprise
Number of Users	Up to 5	Up to 50	Unlimited
Café Locations	1	Up to 3	Unlimited
POS Terminals per Location	1	Multiple	Unlimited
Menu Categories	Drinks, Desserts	Extensible	Fully Customizable

Scalability Feature	Starter	Professional	Enterprise
Database Scaling	Single instance	Shared DB	Distributed / Hosted
Centralized Management	Not available	Limited	Full multi-branch control
Custom Integrations	Not available	Limited API	Full integration support

12. Competitive Analysis and Differentiation

The coffee shop point-of-sale market includes a number of established commercial platforms that offer varying levels of functionality, pricing, and flexibility. This section provides a comparative analysis of the Coffee POS System against leading competitors and identifies the key differentiating factors that position this solution as a strong alternative for cafés seeking a customizable and cost-effective platform.

12.1. Market Overview

The point-of-sale software market for food and beverage businesses is dominated by a small number of well-known commercial platforms. These platforms typically operate on subscription-based pricing models and offer cloud-hosted infrastructure with broad feature sets. While these solutions are widely adopted, they often present challenges for small independent cafés, including high recurring costs, limited customization, and dependency on vendor-controlled infrastructure.

The Coffee POS System addresses these challenges by offering an open-source foundation with optional enterprise support, giving organizations full control over their software environment while remaining accessible to businesses with limited technical resources.

12.2. Competitor Comparison

Feature	Coffee POS System	Square for Restaurants	Toast POS	Lightspeed Restaurant
Licensing Model	Open-source (MIT) with optional commercial support	Freemium with paid tiers	Subscription-based	Subscription-based
Base Cost	Free (open-source); support packages available	Free basic plan; paid plans from ~$60/month	From ~$69/month	From ~$69/month
Source Code Access	Full source code available	Not available	Not available	Not available
Customization	Full customization of menu, features, and UI	Limited configuration options	Limited without developer support	Moderate through add-ons
Self-Hosted Deployment	Supported; on-premise available in Enterprise	Cloud only	Cloud only	Cloud only

Feature	Coffee POS System	Square for Restaurants	Toast POS	Lightspeed Restaurant
Multi-Location Support	Professional and Enterprise packages	Available on paid plans	Available on higher tiers	Available on paid plans
Offline Operation	Supported through local deployment	Limited offline mode	Limited offline mode	Limited offline mode
Technical Support	Optional SLA-based support packages	Community and paid support	24/7 paid support	Email and chat support

12.3. Key Differentiating Factors

The Coffee POS System offers several distinguishing advantages over competing commercial platforms. These differentiators make the system particularly well-suited for independent cafés, growing coffee shop chains, and organizations that prioritize software ownership and flexibility.

Differentiator	Description	Business Benefit
Open-Source Foundation	Distributed under the MIT License, providing full access to source code.	Eliminates vendor lock-in and allows organizations to modify the system to meet specific operational requirements.
Cost Accessibility	The core system is available at no licensing cost, with optional paid support.	Reduces barriers to adoption for small cafés and startups with limited budgets.
On-Premise Deployment	Supports local server and containerized deployment using Docker.	Allows businesses to operate without dependency on external cloud services, improving reliability and data sovereignty.
Modular Architecture	Built on a class-based Java architecture with clearly separated components.	Simplifies maintenance and future feature development without disrupting existing system functionality.
Flexible Commercial Model	Tiered packages (Starter, Professional, Enterprise) align cost with scale.	Allows businesses to adopt only the features and support they require, avoiding unnecessary expenditure.
Full Customization	Menu items, user roles, currency settings, and system configuration are fully customizable through environment variables and the Admin Panel.	Enables cafés to tailor the POS experience precisely to their brand and operational workflow.

12.4. Target Market Positioning

Based on the competitive analysis above, the Coffee POS System is best positioned as a primary solution for the following market segments:

- Independent cafés and small coffee shops seeking a cost-effective POS solution without recurring subscription fees.

- Growing café chains that require multi-location support and centralized management without committing to expensive enterprise software contracts.
- Technology-aware businesses that wish to customize and self-host their POS platform to maintain control over their operational data.
- Franchise organizations that need a scalable and configurable POS foundation that can be adapted across diverse branch environments.

The combination of open-source accessibility, flexible licensing, and structured commercial support packages positions the Coffee POS System as a compelling alternative to subscription-dependent platforms, particularly for organizations that prioritize long-term cost efficiency, software ownership, and operational flexibility.

13. Licensing and Commercial Model

The Coffee Point of Sale System is designed as a scalable solution for coffee shops, cafés, and restaurant environments. To accommodate businesses of different sizes and operational needs, the system is distributed under a flexible licensing and commercial model. This model allows organizations ranging from small single location cafés to large multi-branch coffee chains to adopt the platform according to their operational scale.

13.1. Licensing

The Coffee POS System can be distributed under the **MIT License**, an open-source software license that allows developers and organizations to modify, and distribute the software. Under this license, users are permitted to integrate the software into their own systems, customize the functionality to fit business requirements, and deploy the system in private or commercial environments.

Additionally to the open-source licensing model, organizations using the Coffee POS System may optionally purchase enterprise support services. These services can include system maintenance, software updates, technical troubleshooting, and service-level agreements (SLAs) to ensure reliable operation in production environments. This combination of open licensing and optional commercial support allows the system to remain accessible for small cafés while still providing professional support options for larger businesses.

MIT License Characteristics:

License Feature	Description	Relevance to Coffee POS
Commercial Use	Allows the software to be used in commercial environments without restriction.	Coffee shops and café chains can legally use the POS system in their businesses.
Modification	Users are allowed to modify the source code.	Café owners or developers can customize menu handling, ordering features, or reporting tools.
Distribution	The software can be shared or redistributed.	The system can be installed across multiple café locations or shared between franchise branches.
Private Deployment	Organizations can deploy the system internally without releasing their modifications publicly.	Businesses can run the POS system privately on their own terminals or servers.
Minimal Restrictions	Only requires preserving the original copyright and license notice.	Makes the software easy to adopt and maintain for small development teams or

		startups.

13.2. Commercial Packages

The Coffee POS is offered through a tiered commercial package structure designed to support cafés and coffee shop businesses of different sizes and operational needs. Each package provides a different level of functionality, scalability, and technical support. This structure allows small cafés to implement a simple POS solution while enabling larger coffee shop chains to utilize advanced management capabilities, multi-location support, and dedicated technical assistance.

Starter Package: designed for small cafés or single-location coffee shops
Features include:

- Up to 5 system users (cashiers and administrators)
- 1 café location
- Basic POS order processing interface
- Menu management for Drinks and Desserts
- Basic administrator access through the Admin Panel
- Daily sales tracking and transaction logging
- 30-day order history and audit log
- Standard email technical support
- 99% system availability SLA

Professional Package: For growing coffee shops or small café chains
Features include:

- Up to 50 system users
- Support for up to 3 café locations
- Advanced menu management for Drinks, Desserts, and future menu extensions
- Role-based access for Admins and POS operators
- Extended transaction and order history (1 year)
- Enhanced reporting and sales analytics
- Inventory tracking for common coffee shop ingredients
- Priority email and chat support
- Support for multiple POS terminals within a café
- 99.5% system availability SLA

Enterprise Package: designed for large coffee shop chains, franchise businesses, or organizations requiring high scalability
Features include:

- Unlimited system users
- Unlimited café locations
- Centralized menu and system configuration across all branches
- Advanced analytics and business intelligence reporting
- Unlimited transaction and audit log retention

- Multi-location administrative management tools
- Dedicated 24/7 technical support
- Optional on-premise deployment or private hosting
- Custom feature development and integration options
- 99.9% system availability SLA

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14. Technical Requirement

The technical requirements define the minimum hardware, software, and system environment specifications necessary to successfully deploy and operate the Coffee Point-of-View System. These requirements ensure that the application performs reliably while supporting core functionalities such as order processing, menu management, user authentication, and administrative operations.

14.1. Server Side

The server-side environment of the Coffee Point-of-Sale (POS) System is responsible for supporting the core application logic and ensuring reliable system functionality. Server-side components manage key operations such as user authentication, order processing, menu management, and administrative controls.

Requirement	**Specification**	**Purpose in the Coffee POS**
Java Development Kit (JDK)	JDK 17 or higher	Required to compile and run the Java-based POS application and support the system classes such as POS, Admin, Menu, Drink, and Dessert.
Java Runtime Environment (JRE)	JRE 17 or higher	Provides the runtime environment necessary for executing the POS application on the server or host machine.
Build Management Tool	Maven	Used to manage project dependencies, automate builds, and maintain consistent compilation of the application during development and deployment.
Database System	MySQL or postSQL	Stores persistent data such as menu items, order history, user credentials, and

		administrative information. This allows cafés to maintain transaction records and menu configurations.
Containerization	Docker	Enables consistent deployment across different environments and simplifies installation for café systems or enterprise deployments.

14.2. Client Side

The client-side component of the Coffee POS represents the interface used by café employees to interact with the application. This includes processing customer orders, managing menu selections, and accessing administrative controls through graphical user interfaces.

Requirement	**Specification**	**Purpose in the Coffee POS**
Operating System	Windows 10+, macOS, or Linux	Provides the platform for running the POS desktop application used by café staff to process orders and manage system operations.
Touchscreen Support	POS-compatible touchscreen monitor	Enables faster order entry in café environments where touchscreen POS terminals are commonly used.
Local Network Access	Ethernet or Wi-Fi connection	Allows multiple POS terminals to connect to a shared database or server environment when deployed in larger café setups.
Input Devices	Keyboard and Mouse	Allows employees to interact with the system to enter orders, navigate menus, and perform administrative actions.
Display Resolution	Minimum 1024 × 768	Provides sufficient screen space to display the POS ordering interface, menu items, and administrative management panels.

14.3. Environment Variables

Environment variables allow administrators to configure important system settings without modifying the application source code. These variables control system behavior such as application identification, database connections, administrative access, and POS terminal configuration.

Environment Variable	Description	Purpose in the Coffee POS
APP_NAME	Default value: **CoffeeShopPOS**.	Allows cafés to customize the POS system name displayed on the main interface.
DATABASE_URL	Specifies the database connection string used to connect to the system database.	Enables the POS system to store and retrieve menu items, order history, and user authentication data.
ADMIN_DEFAULT_USER	Defines the default administrator username created during system initialization.	Ensures that the system has an initial administrator account for accessing the **Management Frame**.
ADMIN_DEFAULT_PASSWORD	Defines the initial password for the administrator account.	Allows secure initial access to the system through the **LoginFrame** before custom credentials are configured.
DEFAULT_CURRENCY	Sets the currency displayed in the POS interface (e.g., USD, EUR, TRY).	Ensures prices for **Drinks and Desserts** are displayed correctly for the café's local currency.